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Curiosity on Cutting-Edge Technology via Theory of Planned Behavior and Diffusion of Innovation Theory

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ABSTRACT

Despite the growing importance of cutting-edge technologies, little is still known about the effects of psychological and technical factors on these technologies as associated with consumers' behavioral intentions. Drawing on diffusion of innovation theory (DOI) and theory of planned behavior (TPB), this study investigates the effect of perceived behavioral control, complexity, and compatibility on attitude towards using a smartwatch and behavioral intention. The present study also examines the mediation effect of consumer attitude between the psychological and technical factors of smartwatches and behavioral intention. We have gathered data from 291 actual smartwatch users. Partial-Least-Squares Structural Equation Modelling and Importance-Performance Analysis (IMPA) were used to analyze the data. The results reveal that both psychological and technical attributes play a key role in affecting consumers' behavioral intention, in which attitude mediates this relationship. Moreover, among the predictor variables, compatibility is most significant in terms of explaining behavioral intention to use smartwatches, whereas curiosity and complexity have the highest performance index.

1. Introduction

The consumer information and communications technology industry has been seeing a boom due to an increase in the number of wearable devices, especially with the uptake of smartwatches (Dehghani, 2016; Dwivedi et al., 2022; Kim and Shin, 2015). In the consumer electronics business, wearable devices are a comparatively recent phenomenon. Yet, smartwatches have witnessed exponential growth in recent years. According to Statista's report (2021), annual smartwatch revenues are expected to increase from nine million in 2016 to over 22 million in 2020, owing to their increasing popularity. Moreover, as smartwatch usage is forecast to grow globally in the next few years, the number of watches shipped worldwide has been projected to grow from 37 million units in 2016 to over 253 million units by 2025 (Laricchia, 2022). In particular, the spread of COVID-19 may have prompted consumers to stay at home more, close businesses, and take a furlough, leading to a greater demand for smartwatches and fitness trackers (Osborne, 2021).

There is still very little scientific understanding of the factors affecting a smartwatch's usage intention. Silva and Bonetti (2021) stated that smartwatches are still not very popular among innovative technologies such as smartphones, tablets, and augmented/virtual reality headsets. Furthermore, Adapa et al. (2016), states that factors that affect consumer usage intention regarding smartwatches are yet under-researched. More specifically, Kim and Shin (2015) called for research to reveal other factors that may have a psychological effect influencing smartwatch adoption. Therefore, it is important to pay attention to the psychological aspect as well as the technical aspect to understand the intention to use a smartwatch. Drawing upon this, the current study is scoped to wearable devices' behavioral intention expanding the knowledge of customer experience of smartwatches, covering both psychological and technical aspects. Hence, it investigates the effects of perceived behavioral control, compatibility, complexity, and curiosity on both consumer attitudes towards using a smartwatch and behavioral intention.

Furthermore, for a better and more comprehensive understanding of consumers' behavioral intention to use smartwatches, the present study examines the mediation effect of consumer attitude between the psychological and technical factors of smartwatches and behavioral intention. As the current study is concerned with cutting-edge technology, where behavior can rapidly change due to the emergence of new technolo-

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gies, such mediation analysis is very important as it can explain how customer attitudes intervention influences the behavior it may be expected to change. It clarifies the mechanisms of behavior change in the context of wearable devices. In other words, the change in consumers' adoption intention of smartwatches can be better captured through the influences that may occur between perceived behavioral control, compatibility, complexity, and curiosity and the favorable or unfavorable beliefs that customers develop towards the use of smartwatches. Moreover, there is a lack of knowledge on the mediation role of attitude towards using smartwatches in the existing literature.

Consequently, the research challenges lie not only in expanding the knowledge of smartwatch's usage intention but also in taking the cutting-edge technologies models a further step where psychological factors and technical factors can be considered when consumers' usage intention is explored. As well as the current research challenges the idea of examining collective constructs, by which perceived behavioral control and curiosity have been brought into such context for the first time.

With this comprehensive model, this study aims to fill this gap in the literature by addressing consumers' behavioral intention to use smartwatches. More specifically, this study poses the following research questions:

RQ1: How do perceived behavioral control (derived from Theory Planned Behavior-the psychological aspect) and compatibility and complexity (derived from Diffusion Innovation Theory-technical aspect) impact consumers' behavioral intention to use smartwatches?

RQ2: What is the role of curiosity regarding consumers' behavioral intention to use smartwatches?

RQ3: How does consumer attitudes mediate the relationship between the psychological and technical attributes of smartwatches and consumers' behavioral intention to use smartwatches?

This study provides a new perspective by integrating psychological factors and technical factors with curiosity to understand smartwatch usage intention. The research model is the first to examine these constructs collectively, in which perceived behavioral control and curiosity have been brought into the smartwatch context for the first time. To achieve this, the research model is developed by combining the theory of planned behavior (TPB) (Ajzen, 1991) with the diffusion of innovation theory (DOI) (Rogers, 1995) along with curiosity. The current study, in addition, extends the results using Importance-Performance Analysis (IPMA) by which managerial actions taken by managers and marketers based on both importance and performance can be prioritized.

We structure the remainder of this paper as follows: the first section presents a theoretical background with an extensive literature review on smartwatch studies that are used to establish a conceptual model and then develops research hypotheses that discuss the importance of factors affecting consumers' behavioral intention to use smartwatches. Followingly, the second part includes the data collection procedure, the measures adopted, analysis of the data, and provision of results. The last section summarizes the findings, evaluates implications for scholars and professionals, and presents limitations as well as potential future research.

2. Theoretical background

Consumers' adoption of smartwatches has been examined from different perspectives in the marketing literature. Many theories have been postulated to understand the adoption of smart wearables in the consumer segment. The Theory of Reasoned Action (TRA) (1975; 1980), at the convergence of the consumer behavior and social psychology fields, was used (Ajzen and Fishbein, 1980; Fishbein and Ajzen, 1975). TRA aims to predict consumer intentions and behavior by providing a model to recognize target consumers' attempts at behavioral change (Mathieson, 1991; Sheppard et al., 1988; Wu, 2020). Hartwick and Barki (1994) leveraged the theory to explain information technology adoption in an organization. A third factor was later added to the TRA model: perceived behavioral control. TRA has been used to examine consumers' adoption of smartwatches (Hsiao and Chen, 2018), where it was found that attitude toward using smartwatches affects consumers' purchase intentions.

Hsiao and Chen (2018), using TRA and perceived values, investigated the drivers of smartwatch purchase intention from the hardware, software, design, and value perspectives. Interestingly, they found that interface convenience, perceived content, perceived infrastructure, and design aesthetics significantly influence attitudes toward using a smartwatch. Attitude, in turn, affects purchase intention. Taking a net valence framework perspective, Chuah (2019) observes multiple sequential mediations of smartwatch continuance intention. Chuah (2019) establishes that inspiration and well-being in serial mediates the impact of perceived benefits and previous lifestyle incongruence on continuance intention. However, no impact of perceived risks on continuance intention was recorded.

On the other hand, we can see that Arpaci et al. (2020), Baudier et al. (2020), Baba et al. (2019), Chuah et al. (2016), Choi and Kim (2016), Krey et al. (2016), and Lazaro et al. (2020) used the technology acceptance model (TAM) as a bias for addressing the adoption of a smartwatch. Their results are not much different, as we can find that perceived usefulness and enjoyment affect consumers' attitudes towards smartwatches, and attitude affects consumers' intention. Remarkably, Jeong et al. (2015), employing TAM and perceived similarity for a sample of 326 smartwatch users, found that user experience of smartwatches could affect perceived usefulness and perceived ease-of-use, as well as perceived similarity significantly impacting smartwatches' perceived usefulness, perceived ease-of-use and perceived aesthetics. Perceived usefulness and perceived aesthetics affect acceptance intention. Such results highlight the importance of user experience in the consumer electronics businesses, which was adopted in the Innovation Diffusion Theory (DOI).

Taking a collaboration technology perspective, using a combined model of TAM and DOI Theory, Choe and Noh (2018) address consumers' adoption of smartwatches. With a survey of 1500 South Korean users, Choe and Noh (2018) demonstrate that the compatibility of using a smartwatch positively affects its perceived usefulness. On the other hand, observability and trialability influence smartwatches' perceived ease of use. The same approach was also adopted by Hong et al. (2017), where they combined DOI theory, TAM, expectation confirmation theory, and flow theory. Their results bring new light to the literature as it was established that consumer innovativeness affects both hedonic and utilitarian values. Hedonic and utilitarian values, in turn, influence the continuance intention of smartwatches. Therefore, consumers can purchase smartwatches not only for pleasure, fun, and enjoyment but also for practical uses. Also, Cho et al. (2019) employ stimulus–organism–response theory; studying technological and fashion-related factors find that both characteristics have an impact on product attachment towards smartwatches.

Furthermore, other than TRA, TAM, DOI, expectation confirmation theory, and flow theory, the literature indicates the use of other theories addressing consumers' adoption of smartwatches. For example, Unified Theory of Acceptance and Use of Technology (UTAUT) (Adapa et al., 2016; Beh et al., 2021; Wu et al., 2016), task-technology fit model (Hsiao, 2017; El-Masri et al., 2022), expectation-confirmation model (Nascimento et al., 2018), elaboration likelihood model (Krey et al., 2019), schema incongruity theory (Krey et al., 2019), inertia theory (Ghazali et al., 2020), net valence framework (Chuah, 2019), self-determination theory (Siepmann and Kowalczuk, 2021), innovation resistance theory (Mani and Chouk, 2017) and reactance theory (Ogbanufe and Gerhart, 2022). Such wide use of different theories sug-

gests that considerable uncertainty still exists with regard to the factors that lead to the consumers' adoption of smartwatches and that the adoption of smartwatches can be explained from different perspectives using different factors. Table 1 illustrates the key papers addressing consumers' adoption of smartwatches, stating the research approach, the context, the main theory used, and the key result.

2.1. Conceptual model

The theory of planned behavior (TPB) (Ajzen, 1991) and the diffusion of innovation theory (DOI) (Rogers, 1995) are the key theories that are widely used to predict consumers' adoption and behavior. However, far too little attention has been paid to the control individuals perceive over their inside state or behavior by using a smartwatch. Moreover, the nature of consumer curiosity remains unclear. Even though consumer curiosity is considered a key driver for consumer behavioral intention, there is currently very little known about the role of consumer curiosity in consumers' adoption of smartwatches. In the present study, consequently, we synthesize the above-mentioned theories and consumer curiosity to introduce a research model explaining consumer behavioral intentions towards smartwatches. The constructs, which are examined in this study, are defined in the context of smartwatches, and presented in Table 2.

As smartwatches can be realized in two aspects, the psychological and the technical, the collaboration of TPB and DOI theory can essentially explain consumers' intention to adopt wearable technologies. TPB, exemplified by perceived behavioral control and attitude, can explain the psychological aspect, and DOI theory, represented by compatibility and complexity, can explain the technical aspect. Such a focus on these two aspects, using certain constructs, helps to precisely identify the factors that contribute to consumers' intention to use smartwatches. With the new technologies, individuals are very keen to have control of the technical tasks. Particularly, smartwatches provide a high level of personalized experience to users, which may boost the level of control perception that individuals require. This, therefore, can be reflected in the model using the perceived behavioral control construct, which is defined in the context of smartwatch wearable technologies as the perceived level of control that an individual may have performing tasks related to smartwatches (Cheung and To, 2017). In addition, smartwatches provide services that are associated with other technologies and products individuals use and need to be consistent with the individual habits and current state of my daily life, which highlights the importance of the concept of compatibility in the suggested model. Likewise, there is a substantial necessity for new technologies to be easy to use and understandable. Makanyeza (2017) argues that innovations that are hard to understand or require mental effort are typically rejected. Hence, complexity is considered in the proposed research model. Moreover, as users normally desire to learn about new technologies, we expand the collaboration of TPB and DOI theory by bringing the model curiosity construct. Moreover, as users normally desire to learn about new technologies, we expand the collaboration of TPB and DOI theory by bringing the model curiosity construct. This study expands TPB and DOI theory and is the first to use curiosity in the context of smartwatches. Our theoretical model is presented in Fig. 1.

3. Hypotheses development

3.1. Perceived behavioral control

Perceived behavioral control refers to the perceived effort that an individual may take performing a certain behavior (Cheung and To, 2017). Performing such behavior may be perceived with ease or difficulty depending on experience and expected difficulties (Cheung and To, 2017). Attitude and behavioral intention, on the other hand, are defined as the set of emotions that an individual develops regarding a particular behavior and an indication of an individual's readiness to perform a behavior, respectively (Ajzen and Fishbein, 1980).

According to TPB, consumers' behavioral intention can be predicted through both their attitude and level of perception of behavioral control (Taylor and Todd, 1995). The greater the resources consumers hold, the greater perceived behavioral control they perceive to possess (Elwalda et al., 2016). The more favorable the attitude towards behavior, and the higher the perceived behavioral control, the stronger the intention to perform the behavior (Ajzen and Fishbein, 1980). Perceived behavioral control has been examined in many different models. For example, it was found that perceived behavioral control derived from online customer reviews positively affects both consumers' trust and behavioral intention (Elwalda et al., 2016). It was also established that perceived behavioral control increases are related to behavioral intentions to involve oneself in transactions through social networking services (Hansen et al., 2018).

The logic behind the influence of perceived behavioral control is the motivational influence effect on behavior (Madden et al., 1992). If the perception of control over a certain behavior is low, an individual's intention to perform that behavior is also expected to be low, even if the individual developed a positive attitude (Madden et al., 1992). Hence, we expect that the level of perception of control the consumers possess overusing a smartwatch can affect both attitude and behavioral intention. Similarly, Roberts (1996) contends that perceived behavioral control positively impacts consumers' attitude and purchase intention. Armitage and Talibudeen (2010) and Ajzen and Madden (1986) argue that perceived behavioral control is not an influential construct because it only affects consumers' attitude and intention, but also because it is associated with salient beliefs of the decision-making. Additionally, smartwatch services provide a high level of personalized experience to users, which may raise the level of behavioral control perception. Accordingly, this study proposes that the higher perceived behavioral control, the higher positive attitude and behavioral intention consumers develop. Based on the above discussion, we hypothesize the following:

H1: Perceived behavioral control positively influences (a) consumers' attitude, and (b) the behavioral intention towards smartwatches.

3.2. Compatibility

Compatibility is one of the essential constructs of DOI theory. It refers to the degree to which innovation is perceived to agree with a potential adopter's past experiences, current values, and needs (Kim and Ammeter, 2014). Compatibility can also describe how the novel technology fits in consumers' existing lifestyles and prior experiences (Labrecque et al., 2017). In the smartwatch context, compatibility may refer also to compatibility with other products, such as smartphones and computers (Dehghani, 2018). Studies related to DOI theory reported that compatibility has an influence on the adoption of the smart farm (Yoon et al., 2020), and direct and positive impact on both consumers' attitudes and intention to participate in an online travel community (Agag and El-Masry, 2016). Smartwatches provide services such as real-time messaging that can ease task accomplishment. They can also record the user's daily habits and can be personalized to the user's experience. Such services can raise consumers' behavioral intention to adopt smartwatches (Hsiao, 2017). Compatibility is argued to be a key factor affecting continuous intention to use a smartwatch (Dehghani, 2018). Hsiao (2017), in his study of Apple and non-Apple watches, reports that compatibility positively impacts smartwatch adoption intention. Hence, the following hypothesis is proposed:

H2: Compatibility of using smartwatches positively influences (a) consumers' attitude and (b) the behavioral intention towards smartwatches.

Table 1
Key smartwatch studies paper.

Authors	Research approach	Context (country)	Theoretical basis	Key results
Jeong et al. (2015)	Survey (326 smartphone users)	South Korea	TAM	Perceived usefulness and perceived aesthetics affect consumers' intention of a smartwatch.
Kim & Shin (2015)	Survey (363 smartwatch users)	South Korea	TAM	Cost and attitude affect intention. Perceived uselessness does not affect intention.
Adapa et al. (2016)	Interviews (25 college students and working professionals)	Midwestern	TAM and UTAUT	Availability of fitness apps was a key factor influencing adoption. Usefulness plays an important and consistent role in the adoption of smartwatches.
Chuah et al. (2016)	Survey (226 students)	Malaysia	TAM	Perceived usefulness and visibility affect both attitudes towards using and adoption of smartwatches.
Choi and Kim (2016)	Survey (562)	South Korea	TAM	Self-expressiveness affect intention. Perceived compatibility was a significant predictor of how much people perceive smartwatches to be useful and easy to use.
Jung et al. (2016)	Conjoint survey (123 students)	South Korea	NA	Smartwatch choices are affected by display shape and standalone communication than brand and price.
Kim (2016)	Survey (200 students)	Hong Kong	TAM	A round screen compared with a square screen is more effective in promoting smartwatch adoption. Affective quality, perceived usefulness, and attitude affect intention to use smartwatches.
Krey et al. (2016)	Survey (226 non-users of smartwatches)	Malaysia	TAM	Perceived usefulness and visibility affect smartwatch's adoption.
Mani & Chouk (2017)	Survey (402 students)	France	Innovation Resistance Theory	Perceived uselessness, perceived novelty, perceived price, intrusiveness and Self-efficacy affect consumer resistance to a smartwatch.
Wu et al. (2016)	Survey (212)	Taiwan	DOI, TAM, UTAUT	Relative advantage, result demonstrability, perceived enjoyment, and social influence affect consumers' attitudes toward using. Gender has no significant impact on the acceptance of smartwatches.
Hong et al. (2017)	Survey (276 who had experience using smartwatches)	Taiwan	DOI, TAM, Expectation confirmation theory, and flow theory	Consumer innovativeness affects both hedonic value and utilitarian value. Hedonic value and utilitarian value affect continuance intention.
Hsiao (2017)	Survey (341 potential adopters of smartwatches)	Taiwan	Task-technology fit model, DOI, new product adoption model	Task-technology fit affects adoption intention. Smartwatch tasks affect task-technology fit. Relative advantage and compatibility affect both task-technology fit and adoption of smartwatches.
Choe & Noh (2018)	Survey (1500)	South Korea	TAM, DOI	Observability and trialability affect perceived ease of use. Perceived usefulness affects intention to use smartwatches.
Dehghani (2018)	Qualitative (non-participant netnography; Amazon customers' reviews).	Not specified	NA	Perceived usefulness, perceived ease of use, enabling technologies, functionality, compatibility, fashionability, health motivation, and complementary goods are expected to have an impact on the continuous usage intention of smartwatches.
Dehghani et al. 2018	Survey (383 actual smartwatch users)	Not specified	Technology acceptance theories	Aesthetic appeal, operational imperfection, and hedonic motivation affect the continuous intention of smartwatches.
Hsiao & Chen (2018)	Survey (260 users of the Apple Watch)	Taiwan	TRA and perceived values	Emotional value, performance/quality value, price/value for money affect intention to purchase smartwatches.
Nascimento et al. (2018)	Survey (574 smartwatch users)	USA	Expectation-confirmation model,	Habit, satisfaction, and perceived usefulness affect smartwatches' continuance intention.
Shin & Biocca (2018)	Survey (336 students and nonstudents)	South Korea	NA	Coolness is a social motive for switching a smartwatch. Identity formation is related to smartwatch upgrade intention.
Anggraini et al. (2019)	Survey (116 smartwatch users)	Indonesia	Customer's Perception	Brand and prices influence the perceptions of customers using smartwatches. Customers are less aware of the smartwatch usability.

(*continued on next page*)

Table 1 (*continued*)

Authors	Research approach	Context (country)	Theoretical basis	Key results
Baba et al. (2019)	Survey (501 students)	Malaysia	TAM	Perceived usefulness and perceived ease affect intention to use a smartwatch. There is no cost, privacy, or health risk effect on intention to use a smartwatch.
Chuah (2019)	Survey (324 smartwatch users)	Malaysia	Net valence framework,	There is no impact of perceived risks on the continuance intention of smartwatches.
Dutot et al., 2019	Survey (446 students)	China, France, and Thailand	TAM	The intention to use smartwatches is influenced by a specific number of factors that could vary depending on the country.
Krey et al. (2019)	Survey (999 non-smartwatch users)	Malaysia	Elaboration likelihood model, Schema incongruity theory, TAM	Usefulness, ease of use, enjoyment, visibility, self-expressiveness, and personal innovativeness affect attitude toward using a smartwatch.
Arpaci et al. (2020)	Survey (491 students)	Malaysia	TAM	Perceived usefulness and perceived ease affect smartwatch's behavioral intention.
Baudier et al. (2020)	Survey (1,197 smartwatch owners and users)	USA, UK, Germany, and France	TAM	Perceived usefulness, perceived ease of use, perceived playfulness, and perceived connectivity affect attitude towards using smartwatches.
Ghazali et al. (2020)	Survey (308 non-users of smartwatches)	Not specified	DOI, Inertia theory	Attitudes of the low status quo satisfaction groups are more strongly affected by perceived ease of use and trialability than high status quo satisfaction groups.
Lazaro et al. (2020)	Survey (76 older adults, the majority had no experience using smartwatches)	Not specified	TAM	Social support affects consumers' attitude. Accessibility affects the smartwatch's behavioral intention.
Siepmann &Kowalczuk (2021)	Survey (335 actual smartwatch users	Germany	Self-Determination Theory	Perceived usefulness affects continuance intention. Self-quantification positively affects confirmation. Goal-pursuit motivation has a positive impact on perceived usefulness. There is an effect on device annoyance on satisfaction.
El-Masri et al. (2022)	Survey (300 smartwatch respondents)	Qatar	Task-Technology Fit Theory	TTF has a positive impact on Utilisation. Actual TTF and Perceived TTF positively affect TTF. Moreover, utilisation has a positive impact on satisfaction. Technology identify fit (TIF) positively influences satisfaction.
Ogbanufe &Gerhart (2022)	Survey(151 respondents)	UK	Reactance Theory	Value incongruence and social influence affect smartwatch reactance

Table 2
Constructs with definition

Constructs	Definition	References
Perceived behavioral control	Perceived behavioral control refers to the perceived level of control that an individual may have performing tasks related to smartwatches.	Cheung and To, (2017)
Compatibility	Compatibility refers to using a smartwatch that would be consistent with the individual current state of my daily life, habits and preferences.	Dehghani (2018)
Complexity	Complexity refers to the perceived ease of use of using a smartwatch.	Rogers (2003)
Curiosity	Curiosity refers to the desire to seek and learn new information about smartwatches.	Litman (2010)
Consumer attitudes	Consumer attitudes represent consumers who may develop favorable and unfavorable beliefs towards the use of smartwatches.	Ajzen and Fishbein, (1980); Fishbein and Ajzen, (1975)
Behavioral Intention	Behavioral Intention refers to the perceived likelihood that an individual will engage in using a smartwatch.	Ajzen and Fishbein, (1980); Fishbein and Ajzen, (1975)

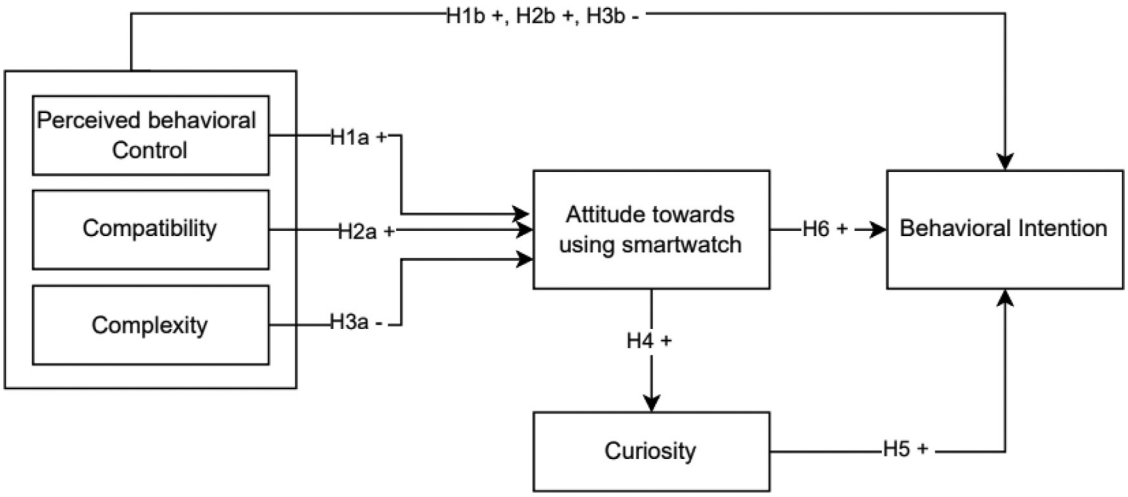

Fig. 1. Proposed research model (Adapted from Sources Ajzen 1991, Rogers 1991)

3.3. Complexity

Complexity is defined as the degree to which an innovation is difficult to understand, and use (Rogers, 2003). It is referred to as the perceived ease of use, as an individual using innovation effortlessly (Puschel et al., 2010). Innovations that are hard to understand or require mental effort are typically rejected (Makanyeza, 2017). For self-handling, high-tech devices such as smartwatches, perceived ease of use is a key predictor for adoption. Possible challenges for adopting smartwatches are the use of higher cognitive inputs (Mukherjee and Hoyer, 2001; Oreg, 2003), small screen (Adapa et al., 2016), and incurring a higher cost as compared to conventional wristwatches (Murray and Häubl, 2007). Therefore, consumers using traditional wristwatches might overestimate the intricacies of using a smartwatch. Smartwatches have easy to use and simple user interfaces, which are easy for users to operate (Hsiao, 2017). The easier the use of smartwatches, the greater the probability consumers will develop a positive attitude towards using them. Prior research has revealed that ease of use has a strong effect on both consumers' attitude and consumer adoption intention of smartwatches (Arpaci et al., 2020; Baba et al., 2019; Baudier et al., 2020; Choe and Noh, 2018). For example, Hsiao (2017) finds that the degree to which a smartwatch is difficult to understand and use negatively affects smartwatch adoption intention. Supporting this, Dehghani (2018) in his conceptual research on wearable technology claim that perceived ease of use of an innovation positively influences the continuous intention to use this innovation. Therefore, we propose the following:

H3: Complexity of using smartwatches negatively influences (a) consumers' attitude and (b) the behavioral intention towards smartwatches.

3.4. Curiosity

Curiosity refers to the desire to seek and learn new information (Litman, 2010). Curiosity is seen as a passion motivational intensity (Loewenstein, 1994). Essentially, curiosity embraces two main objects: first, stimulating positive feelings through gaining information; and second, reducing denial of unwanted information, which is known as deprivation curiosity (Litman, 2010). Previous research has related curiosity with boredom and sensation-seeking (Loewenstein, 1994). The more an individual is motivated, the more he or she desires to seek information. If this passion for information, thus, was preceded by a positive set of emotions that an individual develops regarding a particular behavior, the desire to seek and learn new information will probably be higher. Accordingly, we believe that consumers' curiosity is trigged by the attitude that consumers develop about a certain behavior. The present study, therefore, hypothesizes the following:

H4: Consumers' attitude towards using smartwatches positively influences consumers' curiosity for smartwatches.

In addition, curiosity is found to have a positive effect on both perceived ease of use of virtual reality (Manis and Choi, 2019), and perceived enjoyment of mobile health services (Liu et al., 2019). Additionally, Kashdan et al. (2004) shows that individuals with higher trait curiosity are more expected to draw attention to new and challenging stimuli. Similarly, but in a different context, curiosity is reported to drive consumers' intention to involve themselves with the posted announcement of new products (Gerrath and Biraglia, 2021), and to influence consumers' behavioral intention to select sustainable restaurants (Tommasetti et al., 2018). In this respect, Spielberger and Starr (1994) argue that curiosity addresses individuals' future choices to such an extent that, in many circumstances, their behavior can be de-

termined exclusively by their prior curiosity. As well as, through a field experiment and two laboratory experiments, Yang et al. (2020) reveals that an increase in consumers' curiosity toward an augmented reality advertisement can enhance consumers' attitude toward the ad. Further, consumer curiosity is found to significantly predict purchase motivation (Hill et al. 2016). Consequently, we expect that consumers with high trait curiosity have the desire to seek the experience of learning about smartwatches, and hence are more likely to have an intention to use. Accordingly, the following hypothesis is proposed:

H5: Consumers' curiosity positively influences behavioral intention towards smartwatches.

3.5. Attitude-behavioral intention

According to the TPB, behavioral intention can be greatly determined by attitude (Ajzen and Fishbein, 1980). In the technology context, attitude is also contended to be a key factor affecting consumers' intention to adopt technologies (Park and Chen, 2007). Similar results are found in the context of smartwatches. For example, Hsiao and Chen (2018), using a sample of Apple smartwatch users, reveal that users' attitude toward using smartwatches directly influences their intention to purchase smartwatches. Furthermore, utilizing the technology acceptance model (TAM), Dutot et al., (2019) claim that attitude toward smartwatches positively affects consumers' intention in three different countries, named China, France, and Thailand. Likewise, the intention to adopt smartwatches is found to be positively affected by the attitude toward using smartwatches (Krey et al., 2019). Chuah et al. (2016) also establish that the intention to adopt a smartwatch is positively associated with attitude. Replicating prior studies, the following hypothesis is proposed:

H6: Consumers' attitude positively influences behavioral intention towards smartwatches.

3.6. The mediating role of consumer attitude

Phelps and Hoy (1996) highlight that consumer attitude can relate to both favorable and unfavorable beliefs. In this study, attitudes represent consumers who may develop favorable and unfavorable beliefs towards the use of smartwatches. It is exposed that consumers' adoption intention can be enhanced by perceived behavioral control (Ajzen and Fishbein, 1980; Elwalda et al., 2016), compatibility (Hsiao, 2017), complexity (Cho and Sagynov, 2015; Elwalda et al., 2016), and curiosity (Loewenstein, 1994). Perceived behavioral control, compatibility, complexity, and curiosity (Choe and Noh, 2018; Dutot et al., 2019; Krey et al., 2019; Loewenstein, 1994) are influenced by attitude. Choe and Noh (2018) report that the greater the innovation is perceived to agree with a potential adopter's past experiences, the higher the possibility of using this innovation through the mediation of perceived usefulness. Similarly, Choi and Kim (2016) in their investigation of the factors affecting smartwatch adoption, establish that perceived compatibility affects perceived usefulness, which in turn affects attitude toward using that leads to consumer's behavioral intention. Hong et al. (2017), also reveal that the relationship between consumer innovativeness and continuance intention to use a smartwatch is mediated by utilitarian and hedonic values.

Attitude also is cited to be a key determinant of behavioral intention (Ajzen and Fishbein, 1980; Chuah et al., 2016; Chatterjee et al., 2022). Based on this, we expect consumer attitudes to play a mediating role between the psychological and technical aspects of smartwatches and consumers' behavioral intentions. Accordingly, the current study hypothesizes the following:

H7: Consumers' attitude mediates the relationship between (a) perceived behavioral control, (b) compatibility, (c) complexity, and behavioral intention towards smartwatches.

Table 3
Sample profile.

Variables	Frequency Percentage (%) N=291	
Gender		
Female	132	45.4
Male	158	54.3
Missing	1	0.3
Educational degree		
High School	22	7.5
Bachelor's Degree	169	58.1
Master's Degree	95	32.6
Ph.D.	5	1.7
Age		
24 or younger	73	25.1
25 to 34 years old	68	23.4
35 to 44 years old	90	30.9
45 to 54 years old	39	13.4
55 or above	21	7.2
Income		
Less than $1,000	2	0.7
$1,000 to less than $1,500	22	7.6
$1,500 to less than $2,000	112	38.5
$2,000 to less than $2,500	98	33.7
More than $2,500	46	15.8
Prefer not to say	11	3.8
Price willing to pay for a smartwatch		
$0 to $49.99	23	7.9
$45 to $99.99	65	22.3
$100 to $149.99	84	28.9
$150 to $199.99	107	36.8
Over $200	12	4.1

4. Research methodology

4.1. Sample and data collection

A quantitative survey methodology was used to perform the aims of the study. To verify the research model and conduct hypotheses testing, Data were collected through a survey of smartwatch users. The questionnaire form online was prepared and distributed through the third author to the actual smartwatch users in Brazil. We expressed the aims of the study and informed the participants of data confidentiality before starting the survey. The survey questionnaire form consisted of two parts. In the first part, a set of 28 statements was presented (see Table 3 for details). In the second part, the questionnaire form covered the usual sociodemographic questions to describe the sample profile (age, gender, educational degree, income, and price willing to pay for a smartwatch). First, to validate the questionnaire form, a pilot study, applying to 10 experts, was conducted to reduce uncertainties regarding inconsistencies in language and the clarity of each statement. Based on the feedback given, the questionnaire form was further improved. The questionnaire form was designed and distributed online for users using a smartwatch. Participants were selected via a simple random sampling method. 350 responses were collected. Of those who responded, we could not use 59 responses because of missing values or given wrong responses; for example, not giving a true answer to a paying attention question (Tick Number "3"). Hence, our final sample size was 291 (83% response rate).

Table 3 illustrates an overview of participants' demographic characteristics. The findings illustrate that 54.3% of the participants were male and 45.4% were female. Further, 25.1% of participants were age 24 or younger, 23.4% aged 25-34 years, 30.9% aged 35–44 years, 13.4% aged 45–54 years, and 7.2% were aged 55 years and above. Many participants (90.7%) held at least a bachelor's degree. The wide range of the sample had income mostly between $1,500 to less than $2,500 (72.2%). Regarding the price willing to pay for a smartwatch. 36.8% of respondents stated that they were willing to pay between $150 to $199.99. 28. 9% of respondents stated that they were willing to pay between $100 to

$149.99. 22.3% of respondents stated that they were willing to pay $45 to $99.99.

4.2. Survey instrument and measures

We adapted constructs, namely complexity, perceived behavioral control, compatibility, attitude, curiosity, and behavioral intention, from previous studies (e.g., Al-Jabri and Sohail, 2012, Manis and Choi, 2019; Min et al., 2018; van Dolen et al., 2007; Wu et al., 2016). The scales have been amended to the smartwatch context (e.g. adding "Using a smartwatch to the items"). Likert-type five-point (1 = "strongly disagree", 5 = "strongly agree") multi-item scales were used to measure the constructs. We preferred the 5-Likert scale rather than 7 Likert scales, as it provides easier responding process than 7 Likert scales (Aybek and Toraman, 2022). As well as 5-Likert scale reduces the "frustration level" that the respondents could encounter (e.g., Babakus and Mangold 1992, Sachdev and Verma 2004). A 5- or 7-point scale, moreover, are found to be more expected to deliver an average score higher than the highest possible score achievable, compared to other scales (Dawes, 2008). Table 4 provides all constructs with descriptive information.

4.3. Data analysis

Partial least squares structural equation modeling (PLS-SEM) was applied through SmartPLS 3.2.8 (Ringle et al., 2015) in this study. The PLS-SEM is a "regression-based" and "component-based" approach that helps to lessen the residual variances of the endogenous constructs (Hair et al., 2011), hence it is different to covariance-based structural equation modeling (CB-SEM). Moreover, compared to CB-SEM, PLS-SEM has less strict rules in terms of sample size, measurement scales, and normality assumptions (Chin, 2010; Wong, 2013). A sample size of 291 is sufficient for PLS-SEM, with existing studies recommending that a sample size should be at least 100 (Ling et al., 2021). Additionally, PLS was used in this study because of common usage in technology adoption studies and its capacity to investigate the relationship between the observed and latent variables and the relationship between the exogenous and endogenous latent variables (Hair et al., 2017; Jadil et al., 2022; Sharma et al., 2022). Further, using G*Power analysis, we found that a minimum of 138 samples was enough to obtain a power of 0.95 for analysis (Faul et al., 2009). Hence, the sample size of this study surpasses the required threshold.

The PLS-SEM analysis involves two steps. In the first step, the measurement model, which covers containing convergent and discriminant validity, is examined. To illustrate how the statements are correlated with each other and to understand if the statements are covered in the same measurement, convergent validity was checked through the composite reliability (CR) and the average variance extracted (AVE), as well as by examining factor loadings of individual measures (Hair et al., 2019). Another issue is discriminant validity to explain whether two constructs are statistically different. To ensure the discriminant validity of constructs in the research, three criteria (Fornell and Larcker, HTMT, and cross loadings) were checked. In the second step, the structural model was analyzed, which comprises path coefficients, their p-values, and t-values. Accordingly, in this step, it is stated that the hypotheses are supported or rejected.

Table 4
Descriptive of the measurement items.

Measurement Items 1 = Strongly disagree, 5 = Strongly agree	**Mean**	**St. Dev.**
Compatibility (Al-Jabri& Sohail, 2012)		
Using a smartwatch would be consistent with my current preference	3.43	1.049
Using a smartwatch would be consistent with my current habits	3.41	1.070
Using a smartwatch would be consistent with the current state of my daily life	3.31	1.076
Using a smartwatch would match my living experience	3.25	1.087
Using a smartwatch would be consistent with every aspect of my daily life	3.15	1.139
Complexity (Min et al., 2018)		
Using a smartwatch does NOT require technical skills	3.84	1.083
Using a smartwatch does NOT require a lot of mental effort	3.25	1.087
Using a smartwatch would NOT be frustrating	3.34	1.144
Using a smartwatch would NOT be a challenge	3.78	1.139
Interacting with a smartwatch would NOT confuse me	3.26	1.081
Smartwatch interface looks easy to understand	4.04	1.102
Curiosity (e.g., Exploratory Information Seeking, Manis & Choi, 2019)		
I like to shop around and look at displays	4.19	0.908
I often read advertisements just out of curiosity	3.52	1.118
Reading mail advertising to find out what's new is a waste of time	2.83	1.084
I like to browse through catalogs or online stores even when I don't plan to buy anything	3.49	1.199
Perceived behavioral control (van Dolen et al., 2017)		
I feel much control over my communications social media and me	3.25	1.062
Through using a smartwatch, I have a direct influence on getting	3.26	1.081
Smartwatch enables to get a grip on the necessary information	3.40	1.123
Smartwatch will give me more control over my communications	3.30	1.121
Attitude (Wu et al., 2016)		
Using a smartwatch would be a positive decision	3.88	0.920
Using a smartwatch would be a smart decision to make	3.74	0.989
I have a positive impression of using a smartwatch for work	3.32	1.081
I would feel excited to purchase a smartwatch	3.45	1.083
I would be happy to use a smartwatch	3.51	1.061
Behavioral Intention (Wu et al., 2016)		
I would be willing to use a smartwatch	3.87	0.962
I would be willing to use a smartwatch if I possessed one	4.07	0.893
I would be willing to let a smartwatch help me to perform different kinds of tasks	3.53 3.72	1.001
I intend to use a smartwatch in the near future		1.067

5. Results

5.1. Measurement model evaluation

Table 5 demonstrates the factor loadings, scale reliability, and validity measures. Some statements from curiosity and complexity were eliminated because of low loadings. The other item loadings on respective constructs (compatibility, complexity, curiosity, perceived behavioral control, attitude, behavioral intention) were above 0.70, which is the suggested minimum value (Hair et al., 2019). In terms of composite reliability, each variable was higher than 0.8, which also met the minimum requirement of 0.70 as an acceptable threshold (Nunnally et al., 1967). The AVE values for all constructs were bigger than the lowest suggested value of 0.50, indicating that the items satisfied convergent validity (Bagozzi and Yi, 1988). Further, Cronbach's alpha and rho_A were greater than 0.70, as suggested (Ali et al., 2017; Nunnally and Bernstein, 1994).

Discriminant validity was evaluated with three criteria that compromise Fornell and Larcker's, HTMT and cross-loading (Hair et al., 2017). First, Fornell and Larcker's criterion is assessed by comparing the square roots of AVE with correlations among latent constructs (Fornell and Larcker, 1981). The square root of AVE for the construct was higher than the other constructs' square correlation, which showed that discrimi-

Table 5
Reliability and validity of measurement model.

	Loadings
Compatibility (α=0.940, CR= 0.954, AVE=0.806, Rho_A =0.941)	
COM1	0.882
COM2	0.928
COM3	0.911
COM4	0.885
COM5	0.882
Complexity (α=0.860, CR= 0.906 AVE=0.706, Rho_A =0.864)	
CLX1	0.771
CLX2	0.921
CLX3	0.895
Curiosity (α=0.742, CR= 0.853, AVE=0.660, Rho_A=0.744)	
CUR1	0.783
CUR2	0.811
CUR3	0.842
Perceived behavioral control (α=0.929, CR= 0.949, AVE=0.824, Rho_A=0.931)	0.883
PBC1	0.917
PBC2	0.908
PBC3	0.923
PBC4	
Attitude (α=0.891, CR= 0.920, AVE=0.698, Rho_A=0.901)	
AT1	0.878
AT2	0.875
AT3	0.721
AT4	0.853
AT5	0.840
Behavioral Intention (α=0.847, CR= 0.898, AVE=0.690, Rho_A=0.874)	0.923
BI1	0.897
BI2	0.724
BI3	0.760
BI4	

[Note: α = Cronbach's Alpha; CR = Composite reliability; AVE = Average Variance Extracted: Rho_A = Reliability coefficient; AT: Attitude, BI: Behavioral Intention, COM: Compatibility, CLX: Complexity, CUR: Curiosity, PBC: Perceived Behavioral Control]

Table 6
Discriminant Validity (Fornell Larcker Criterion).

	AT	BI	COM	CLX	PBC	CUR
AT	**0.835**					
BI	0.794	**0.815**				
COM	0.650	0.578	**0.898**			
CLX	0.634	0.602	0.778	**0.840**		
PBC	0.547	0.448	0.531	0.571	**0.908**	
CUR	0.375	0.396	0.436	0.436	0.366	**0.812**

[Note: AT: Attitude, BI: Behavioral Intention, COM: Compatibility, CLX: Complexity, CUR: Curiosity, PBC: Perceived Behavioral Control]. *Bold figures are square root of average variance extracted (AVE) Figures below the AVE line are the correlations between the factors*

nant validity had been achieved. The square roots of all AVEs range from 0.812 to 0.908. All values are provided in Table 6.

HTMT is another criterion with which to assess discriminant analysis. It is "an estimate of what true correlation between two constructs would be if they perfectly measure" (Hair et al., 2017). To achieve this criterion, the HTMT ratio should be less than 1.00 (e.g., Ogbeibu et al., 2018). Table 7 shows that since all values met the threshold, discriminant validity had been achieved.

Since the model fit is largely based on the covariance-based matrix, partial least squares do not exactly predict the goodness of fit. Yet, SmartPLS 3 includes the SRMR as model fit criterion for PLS path modeling, which are recommended to be smaller than 0.08 (Hu and Bentler, 1999). The SRMR for our final structural model was 0.063, which shows an acceptable model fit.

5.2. Structural model evaluation

The structural model was evaluated through the PLS approach to structural equation modeling. The PLS algorithm and the bootstrapping resampling method, with 291 cases and 5000 resamples, were utilized to evaluate the structural model. The findings of the model estimation, covering standardized path coefficients, the significance of the paths, the value of the *t*-test is reported in the Table 9. We also checked the amount of variance explained (R^2), and cross-validated redundancy (Q^2). The values of R^2 for attitude, curiosity, and behavioral intention were 0.499, 0.140, and 0.713, respectively. Cross-validated redundancy (Q^2) is another significant criterion with which to predict model accuracy. The Stone-Geisser's Q^2 values are determined through a blindfolding procedure to evaluate the predictive relevance of the exogenous constructs on endogenous constructs (Merli et al., 2019). In this study, the values of (Q^2) for attitude, curiosity and behavioral intention were 0.342, 0.087 and 0.482, respectively.

We also applied a multicollinearity check of the latent constructs (Kock, 2015), which is one of the methods by which one can assess common method bias. As highly recommended by Kock (2015), the inter-construct VIFs (variance inflation factor) should be less than '5'. Table 8 demonstrates the results of VIF values, varying from '1' (lowest value) to '2.907' (largest value). All the values are less than '5', which is the minimum acceptance value. Therefore, multicollinearity is not a serious issue in this study.

In support of Hypothesis 1a, perceived behavioral control has a significant positive impact on attitude (b = 0.229, t = 4.835, p < 0.001). Hypothesis 2a states that compatibility is positively related to attitude. This hypothesis was also supported (b = 0.347, t = 4.646 p < 0.001). Hypothesis 3a states that the complexity is positively related to attitude. This hypothesis was also supported (b = 0.233, t = 2.963, p < 0.05). On the other hand, Hypothesis 1b stated that perceived behavioral control has a positive impact on behavioral intention; but H1b are not statistically supported (b = -0.067, t = 1.301, p > 0.05). Hypothesis 2b states that compatibility have a positive impact on behavioral intention. However, behavioral intention is not statistically explained by compatibility (b = -0.336, t = 0.626, p < 0.266), hence H2b is not supported. Nevertheless, in support of Hypothesis 3b (b = 0.142, t = 2.348, p < 0.05), complexity has a significant impact on behavioral intention. In support of Hypothesis 4 and As per Hypothesis 6, attitude has a significant positive impact on curiosity (b = 0.375, t = 7.863, p < 0.001) and behavioral intention (b = 0.798, t = 24.123, p < 0.001), respectively. Finally, Hypothesis 5 states that curiosity is positively related to behavioral intention. This hypothesis was also supported (b = 0.096, t = 2.154, p < 0.05). The validated proposed model is presented in Fig. 2.

In the same way, the effect size is utilized to anticipate the degree of the relationship between the latent variables. Lowry and Gaskin (2014) have stated that the effect sizes of 0.35, 0.15, and 0.02 display a large, medium, and small effect for each construct. As highlighted in Table 9, the model has large, medium, and small effect sizes.

5.3. Testing mediation

For a better and comprehensive understating of consumers' behavioral intention to use smartwatches, we also tested whether attitude and curiosity in this research performed a mediating role on the relationship between the independent variables (complexity, perceived behavioral control, and compatibility) and behavioral intention. To do so, the bootstrapping method was applied as suggested by Zhao et al. (2010) and the results are provided in Table 10. Hair et al. (2021) put forward that when the indirect effects are statistically significant, the mediation role can be built. Based on this, we found that all indirect effects were significant; however, since there is no direct relationship between compatibility and behavioral intention and between perceived behavioral control and behavioral intention, we cannot support the mediation role of attitude in those relationships.

Table 7
Discriminant Validity (Heterotrait-Monotrait Ratio-HTMT) Test and Model Fit.

	AT	BI	COM	CLX	PBC	CUR	Items	Saturated Model
AT							SRMR	0.063
BI	0.947						NFI	0.825
COM	0.704	0.639						
CLX	0.733	0.711	0.844					
PBC	0.601	0.498	0.569	0.657				
CUR	0.456	0.493	0.521	0.588	0.44			

[Note: AT: Attitude, BI: Behavioral Intention, COM: Compatibility, CLX: Complexity, CUR: Curiosity, PBC: Perceived Behavioral Control]

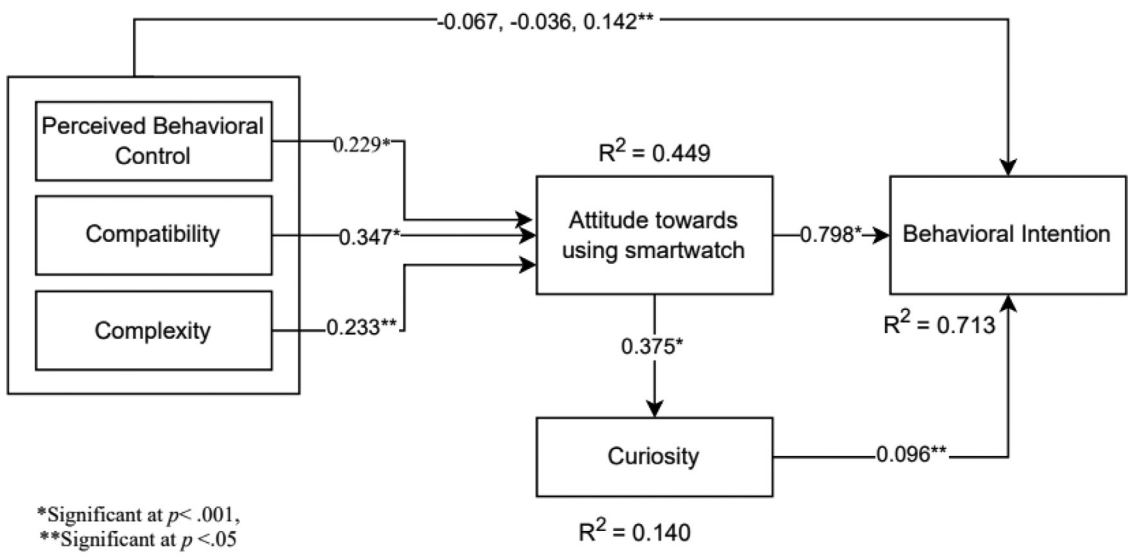

Fig. 2. Validated research model.

Table 8
Common method variance test: inter-construct collinearity (VIFs).

	Attitude	Behavioral Intention
Attitude	-	2.004
Compatibility	2.599	2.885
Complexity	2.761	2.907
Perceived behavioral control	1.522	1.649
Curiosity	1.000	1.301
Behavioral Intention	Endogenous Variable	

5.4. Importance-performance (IPMA) analysis

Besides bootstrapping with 5000 subsamples, proving that some of the path coefficients were statistically significant, we also applied importance-performance map analysis (IPMA) to determine potential areas of improvement in behavioral intention towards smartwatches. IPMA is a procedure that improves regression analysis or path modeling through an index value-based assessment. IPMA provides insights for scholars by stating the areas of low and high performance based on the findings of PLS-SEM (Hock et al., 2010; Ogunmokun et al., 2020). To achieve IPMA, some requirements were met. First, the rescaling of the latent variable scores should vary from 0 to 100 and all the scales should be either metric or quasi-metric (Ringle and Sarstedt, 2016; Sarstedt and Mooi, 2014). Secondly, all scales in the study should have the same coding (from a lower value to the highest value) (Ringle and Sarstedt, 2016). The other requirement is that the outer weights for all the scales should be positive. Finally, the variance inflation factor (VIF) should be checked so that it does not create a collinearity issue. As suggested by Hair et al. (2014), all indicators should be less than 5, showing there is no collinearity issue. By meeting all these requirements, the IMPA analysis was successfully performed in this research.

It is clear from the IPMA findings that to predict behavioral intentions to use smartwatches, attitude has the highest importance (0.79), followed by compatibility (0.24), complexity (0.16), perceived behavioral control (0.15), and curiosity (0.10). In contrast, in terms of the performance of these variables to predict users' behavioral intentions to use smartwatches, curiosity (69.096) tops the list, followed by complexity (68.772), attitude (65.205), compatibility (57.827), and perceived behavioral control (57.568) seen in Table 11. These findings inferred that while attitude is the most important predictor of consumers' behavioral intentions towards using smartwatches, curiosity should be considered the highest priority (Fig. 3).

6. Discussion

Considering the importance of promising wearable technology devices in people's daily lives, this study focused on specific factors that affect users' behavioral intention towards smartwatches. We categorized compatibility and complexity as the technical perspective and perceived behavioral control, and curiosity as the psychological perspective. Drawing on theory planned behavior (TPB) and diffusion of innovation theory (DOI), we hypothesized that psychological and technical aspects of smartwatches have an influence in both attitudes towards smartwatches

Table 9
Structural model path analysis.

	Original Sample (O)	Sample Mean (M)	Standard Deviation (STDEV)	T statistics	p-values	Effect size (f)
H1a: Perceived Behavioral Control → Attitude	0.229	0.230	0.047	4.835	0.000*	0.069
H1b: Perceived Behavioral Control → Behavioral Intention	-0.067	-0.063	0.051	1.301	0.097	0.009
H2a: Compatibility → Attitude	0.347	0.346	0.073	4.745	0.000*	0.092
H2b: Compatibility → Behavioral Intention	-0.036	-0.038	0.058	0.626	0.266	0.002
H3a: Complexity → Attitude	0.233	0.233	0.079	2.963	0.002**	0.039
H3b: Complexity → Behavioral Intention	0.142	0.140	0.061	2.348	0.009**	0.024
H4: Attitude → Curiosity	0.375	0.379	0.048	7.813	0.000*	0.027
H5: Curiosity → Behavioral Intention	0.096	0.097	0.045	2.154	0.016**	0.027
H6: Attitude → Behavioral Intention	0.798	0.798	0.033	24.123	0.000*	1.852

[Note: *Significant at $p < 0.001$, **Significant at $p < 0.05$]

Table 10
Mediation test results.

	Indirect effects	**t-value**	**p-values**	**Mediation Effect**
Compatibility→ Attitude → Behavioral Intention	0.268	4.493	0.000	No
Complexity → Attitude → Behavioral Intention	0.180	3.038	0.002	Yes
Perceived Behavioral Control → Attitude → Behavioral Intention	0.170	4.701	0.000	No
Attitude → Curiosity → Behavioral Intention	0.032	1.972	0.050	Yes
Compatibility → Attitude → Curiosity	0.130	3.826	0.000	Yes
Complexity → Attitude → Curiosity	0.088	2.750	0.003	Yes
Perceived Behavioral Control → Attitude → Curiosity	0.086	3.922	0.000	Yes

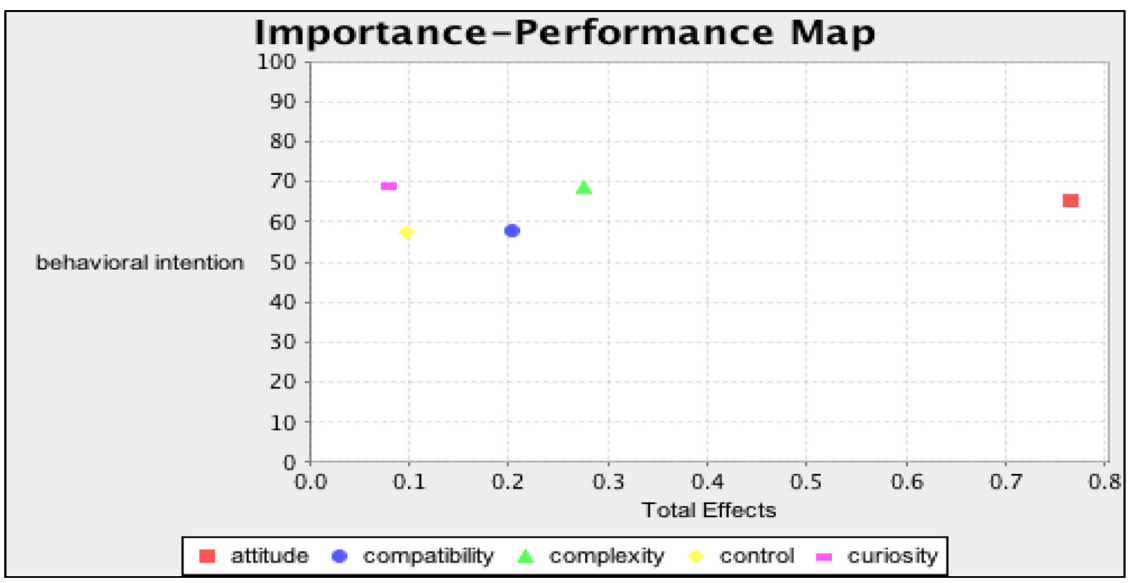

Fig. 3. IMPA factors affecting behavioral intention.

Table 11
Importance-performance index results.

Latent Variables	Behavioral Intention	
	Importance (Total Effect)	Performances (Index value)
Attitude	0.794	65.205
Compatibility	0.242	57.827
Complexity	0.162	68.772
PCB	0.156	57.568
Curiosity	0.091	69.098

and users' behavioral intention towards smartwatches. This study also advances theory by combining TPB and DOI to include the construct of curiosity.

Based on investigating consumers' behavioral intention towards smartwatches, this study analyzed the mediating role of attitude and curiosity on the relationship between latent variables and behavioral intention. Finally, this study is the first in this context to apply importance-performance analysis (IMPA) to identify the determinants that both provide an impact and illustrate the possibility for enhancement.

First, in this study, perceived behavioral control has a positive impact on attitude towards smartwatches. This shows that users who believe in having perceived behavioral control over smartwatches will have positive attitudes towards them. Since smartwatch services enable users to personalize, which might increase the feeling of perceived behavioral control perception, this finding is important. Even though researchers are paying attention to the effect of perceived behavioral control on attitude in different contexts, this is the first study to examine the effects of perceived behavioral control on attitude and behavioral intention in the smartwatch context. On the contrary, previous studies on different contexts found that perceived behavioral control has a significant and direct

effect on behavioral intention (e.g., Jafarkarimi et al., 2016), whereas we found its effect is mediated by attitude in the smartwatch context.

The effect of compatibility on attitude towards using a smartwatch is supported in this study. This finding is consistent with Ghazali et al.'s (2020) and Dehghani's (2018) studies. However, Wu et al. (2016) found that there is a non-significant relationship between compatibility and attitude towards using smartwatches. Similarly, we found that there is a non-significant relationship between compatibility and behavioral intention. This might be explained that when users face huge changes in their daily lives, they can resist using smartwatches (Wu et al., 2016). Like in the relationship between perceived behavioral control and behavioral intention, attitude is the mechanism that can completely clarify the relationship between compatibility and behavioral intention.

Complexity's impact on both attitude and behavioral intention is supported in this research. These findings reflect that since users do not find it complex to use a smartwatch, they tend to have positive attitudes and think of using them in the future. Similarly, Hsiao (2017) found a negative relationship between complexity and smartwatch adoption intention. That is, if the users find it complex to use the smartwatch, they may not tend to adopt them for use. Additionally, the attitude has a positive impact on behavioral intention. This demonstrates that the more positive the attitude towards using smartwatches, the more possible it is that users have a high behavioral intention (Ghazali et al., 2020). This result confirms previous studies' findings in different contexts (e.g., Chuah et al. 2016, Kim and Shin, 2015, Rauschnabel and Ro 2016, Wu et al., 2016).

Another interesting finding is that the effect of attitude on curiosity is supported. This is the first study to examine the relationship between attitude towards using smartwatches and curiosity. The more users have a positive attitude towards using a smartwatch, the more their curiosity to using one. This study is the first endeavor to combine the curiosity literature with the TPB and DOI literature. As expected, the finding shows that curiosity leads to behavioral intentions regarding the smartwatch. In other words, if a user is curious about how to use a smartwatch, or how to work, they are more likely to have behavioral intentions regarding their use. Manis and Choi (2019) have also suggested that curiosity should be used as an antecedent to the Technology Acceptance Model (TAM) in different innovative products. Even though we did not use it as an antecedent of TAM, we used curiosity as an antecedent of TPB in the smartwatch context.

Importance-performance analysis (IMPA) was employed to determine the factors providing an influence and showing possible development. The results show that attitude is of the greatest importance in terms of predicting behavioral intentions to use a smartwatch. Among the other variables, compatibility also has further importance in terms of explaining behavioral intention to use a smartwatch. Curiosity and complexity have the highest performance indexes among the various predictor variables. This finding shows that these two factors should be considered a priority to explain the behavioral intentions of users.

6.1. Theoretical implications

Even though the first smartwatch had fewer features and functions than the current ones, they have been still considered noteworthy cutting-edge technology devices because of the amazing array of functions they offered. Hence, investigating the factors affecting attitudes towards smartwatches and behavioral intentions of smartwatches is still quite important. In spite of the increasing importance of cutting-edge technologies (e.g., Augmented Reality, Blockchain, Artificial Intelligence, Smartwatches, 3D Gaming), little is still known about the effects of psychological and technical factors on these technologies as related to consumers' behavioral intentions. In the light of this importance, this study contributes to the existing literature by providing a new understanding based on technical and psychological factors. The effects of perceived behavioral control, complexity, compatibility, and curiosity on attitude towards using a smartwatch and behavioral intention were captured, providing a better understanding of consumer behavior towards cutting-edge technologies. The current study is the first to examine these constructs collectively, in which perceived behavioral control and curiosity have been brought into the smartwatch context for the first time. The results reveal that both psychological and technical attributes play a key role in affecting consumers' behavioral intention, in which attitude mediates such relationship.

Another major theoretical contribution is combining DOI and TPB. Even though DOI and TPB are separately examined in the smartwatch context, this is the first study to combine significant theories to investigate attitudes toward smartwatches and behavioral intention. Moreover, these theories also integrated with the construct of curiosity, which remains underexamined to answer the paper's research question. The results explain that consumers' attitude towards using smartwatches positively influences consumers' curiosity for smartwatches. Consumers' curiosity, in turn, positively impact behavioral intention towards smartwatches.

As another contribution, this is the first study attempt to apply importance-performance analysis (IMPA) to determine potential areas of improvement in behavioral intention towards smartwatches. In the importance analysis part, attitude is of the greatest significance in terms of predicting behavioral intentions to use a smartwatch. Compatibility also has further importance in explaining behavioral intention to use a smartwatch. Following, curiosity and complexity have the highest performance indexes among the predictor's variables. These results demonstrate that curiosity and complexity should be considered a priority in explaining users' behavioral intentions.

6.2. Practical implications

The findings of this research have implications for smartwatch developers and marketers. Our findings reflect that the technical aspect (compatibility and complexity) and psychological aspect (perceived behavioral control and curiosity) are the antecedents of attitude towards using smartwatches and behavioral intentions towards the smartwatch. Therefore, smartwatch brands or companies should pay attention to these determinants and enhance the user experience of their products if they want to make more money or increase customer-based brand equity (e.g., brand loyalty, and satisfaction).

A lack of complexity in the smartwatch results in its users having a positive attitude, and positively affects behavioral intention towards them. Apparently, users would prefer to use the smartwatch, which is not complex. To support this claim, based on the interview performed by Canhoto and Arp (2017), it was emphasized that users are not keen on smartwatch usage due to their technical limitations. The designer or producer of smartwatches should hence take this finding into consideration to reach an increasing number of users.

Another important determinant of the use of a smartwatch is perceived behavioral control. Since the smartwatch is a personalized innovation device, users might have some concerns about the perceived behavioral control of their information, which affects their thoughts regarding the use of smartwatches. Our study illustrates that when perceived behavioral control is provided by a smartwatch, this will have a positive impact on users' attitudes towards the use of smartwatches. In this case, managers should provide smartwatches that focus more on control-based features. We found that even though there is no relationship between perceived behavioral control and behavioral intention, managers should build this positive relationship by creating positive attitudes towards the use of smartwatches. This means that companies or brands should place a greater emphasis on the control-based feature of smartwatches to create a direct attitude towards their usage. Additionally, as with perceived behavioral control, compatibility is also another important factor in building a positive attitude towards smartwatch usage, though only indirectly. Thus, we recommend that smartwatch brands should design smartwatches that fit with

users' daily lives (e.g., ability to simultaneously use them with their phones).

Another important novel factor affecting smartwatch managers is curiosity. The smartwatch is a still a new, innovative product; therefore, managers should create a feeling of curiosity regarding smartwatch products in different areas such as social media platforms (YouTube ads, Instagram ads, or technological platforms). Based on our findings, curiosity about smartwatches may make users want to use them in the future. In short, managers can benefit from our findings to improve users' experiences with smartwatches, or to reach more users.

6.3. Limitations and future research directions

In this study, we only examined the behavioral intentions towards smartwatches from the psychological and technical perspectives. Future research may look at the different perspectives such as the hedonic perspective (e.g., enjoyment) or the social perspective (e.g., as a sign of high status). From another perspective, Mani and Chouk (2017) have also suggested that privacy and intrusiveness might be examined in the smartwatch context in greater depth. These constructs might be also taken into consideration with the existing model. Additionally, we used a limited number of variables derived from two important theories (TPB and DOI); however, future research might develop a more comprehensive model through the use of these existing variables. Research into perceived behavioral control and curiosity needs further attention in this context. Future research may also consider the effects of these constructs on continuance intention or adoption of the usage of smartwatches by using different mediators such as satisfaction (Kar, 2021). Another important point is that, in this study, a survey instrument was used to gather quantitative data to test the theoretical models (Rana et al., 2022). Even though this is not a limitation, future research may combine semi-instructed interview or in-depth interview data to delve into the smartwatch context. Furthermore, future research can also benefit from different techniques such as the neural network model (e.g., Kitsios et al., 2022) by integrating structural equation modeling to understand the importance of the existing antecedents deeper in cutting-edge technology. In particular, since curiosity is a new variable to test empirically in this context, curiosity should be investigated via different methodological approaches such as through experimental design or longitudinal study. Also, the future researchers should also do some more detailed literature review, weight and meta-analysis (Agarwal et al., 2017; Alalwan et al., 2016; Chhonker et al., 2018; Dwivedi et al., 2022; 2021; Ismagilova et al., 2020; Jadil et al., 2021; Janssen et al., 2019; Patil et al., 2017; Tamilmani et al., 2018) of the factors influencing intentions to use advanced and emerging technologies.

7. Conclusions

Many individuals benefit from smart technology developments such as wearable technology (Dehghani, 2018), especially smartwatches (Ogbanufe and Gerhart, 2022). These technologies, taking place under cutting-edge technologies, play an important role in IS literature. Hence, understanding different factors affecting users' behaviors toward cutting-edge technologies is necessary. Considering this importance, our research has first created a theoretical model drawing on the Diffusion of Innovation Theory and Theory of Planned Behavior with the inclusion of the curiosity construct. Our study investigated smartwatches' behavioral intention from not only a technical aspect but also a psychological aspect. Thus, smartwatch brands and other cutting-edge technology managers can benefit from our findings to understand which determinants ought to be used to provoke users' behavioral intentions. Our study has included the construct of curiosity to understand what curiosity points to technology. This has also yielded a significant research avenue for future research. Another precious point in our study is to be the first study examining the Importance-Performance Analysis. This enables us to compare the factors affecting users' smartwatch behavioral intention. Our conceptual model is expected to make theoretical and managerial contributions by providing novel research streams.

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